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DE LISBOA

NAVISense Vest: Intelligent Multisensory Assistive Vest for Safe Navigation and Awareness

Project Proposal

Problem definition

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People with visual impairments and reduced mobility face challenges navigating safely and independently in complex environments.

Current solutions provide limited information, increasing the risk of collisions and disorientation.

There is a need for a smart wearable solution that improves safety, spatial awareness, and user autonomy.

Solution Beneficiaries

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People with Total or Partial Visual Impairment

Helps users navigate more safely, detect obstacles, and increase independence in daily life.

Older Adults with Mobility Difficulties

Supports orientation and movement, helping prevent falls and increasing confidence

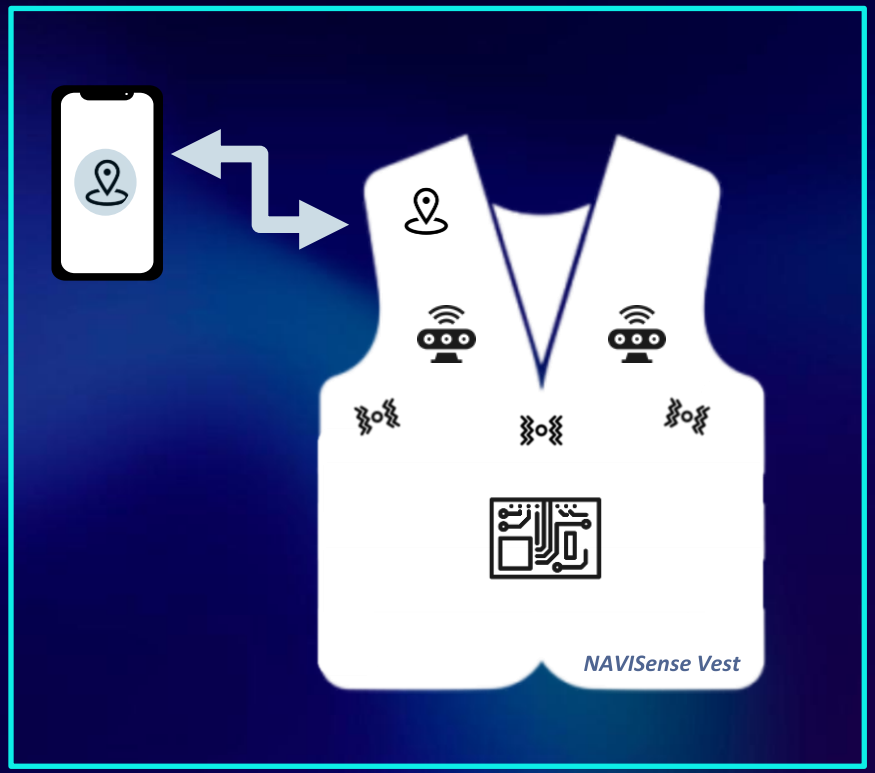
Workers in High-Risk Industrial Environments

Assists navigation in low-visibility areas, improving workplace safety

Rescue and Emergency Teams

Improves spatial awareness and increases safety during emergency operations

Technological Solution



Depth Sensors detect nearby objects and generate spatial distance data.

The vibration motors generates haptic feedback to inform the user about surrounding obstacles and their proximity.

The GPS Module provides real-time user localization and route tracking.

Processing and communication components (ESP32, Raspberry Pi and auxiliary sensors) handle data processing, wireless communication, and system control.

Team & Roles

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Francisco Mariquitos

4G integration, Public Relations and Data Transmission

Raquel Barroso

Vest Design, Sensor choice and Sensor Position

Daniel Khom'yak

Website Design, Dashboard Design and Dashboard Elements

Prof. Francisco Alegria

Mentor

David Reimer

Demonstration Video, Data Transmission and Data acquisition

Tiago Pinto

Pitch Deck, Vest Design and Poster Design

Frederico Pinto

Dashboard Programming, Dashboard Design and Dashboard Elements

Market Opportunities

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- **Relevant target market:**

More than 2.2 billion people live with some form of visual impairment, creating strong demand for navigation and safety solutions.

- **Economic impact:**

Productivity loss associated with visual impairment is estimated at around US$ 411 billion per year, highlighting the potential for assistive technologies.

- **Wearable market growth:**

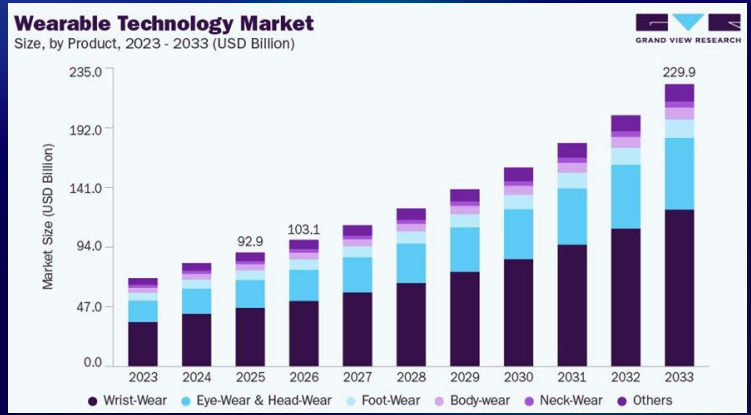
The global wearable technology market is valued at US$ 92.9 billion (2025) and is projected to reach US$ 230 billion by 2033 (≈12% annual growth).

- **B2B safety opportunity:**

Workplace safety wearables represent about US$ 4.0 billion (2024) and are expected to grow to US$ 9.4 billion by 2030.

- **Scalable market potential:**

Applicable across multiple sectors: visual assistance, industrial safety, emergency response teams, and remote safety monitoring.



2.2 billion people

with a near or distance vision impairment

295 million people

have moderate to severe visual impairment

43 million people
have blindness

Technical Challenges

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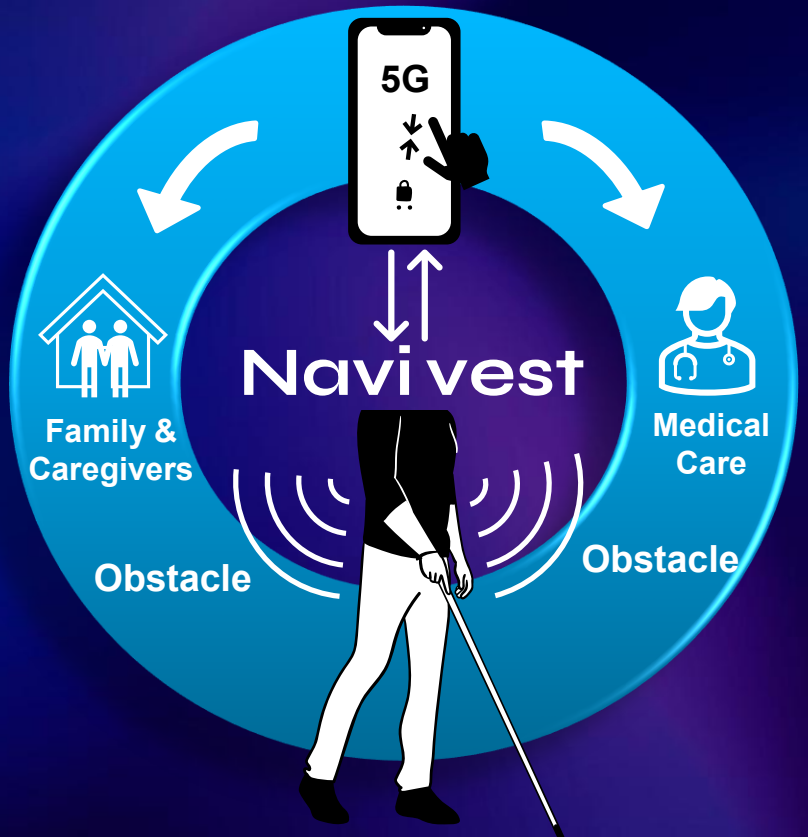
Multiple sensors and vibration motors consume significant energy.

Developing a wearable navigation system with multiple sensors, wireless connectivity, and real-time feedback while keeping the overall cost extremely low is a major challenge.

Bluetooth and 4G connection between vest -> smartphone -> satellite can drop due to interference or user movement.

Low-cost ultrasonic sensors can struggle with reflective surfaces, ambient noise, and angled obstacles.

Solution Requirements



The system must detect obstacles reliably within a defined range (e.g., 20 - 500 cm) using low-cost sensors.

The vest must communicate with a smartphone via Bluetooth, and the smartphone must forward data to a cloud platform via 4G.

Remote dashboards must display live device status, sensor readings, and alerts with minimal latency.

All components and materials must remain within a strict low-cost budget while maintaining system reliability.

Testing and Validation Metrics

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Sensor Performance:

- Detection Range (20-500 cm)
- Distance Accuracy
- Field of View
- Testing of Different Surfaces

Power & Hardware:

- Battery Life Duration
- Power Consuption
- Thermal Stability
- Restart & Recovery Behavior

App & Website Testing:

- Live Data Display
- Update Speed
- Logging Allert Accuracy
- Multi-device & Server Load Testing

System Performance:

- End-to-end Latency
- Transmission Delay
- Delay under different WiFi conditions
- Update Rate

Communication & Data:

- Packet Loss
- Data Transmission Delay
- Buffering Capability
- Data Integrity & Corruption Check

Safety & Reliability:

- False Positive/Negative Rate
- Directional Alert Accuracy
- Environmental Testing
- Testing of Different Surfaces
- System Stability

Project Timeline

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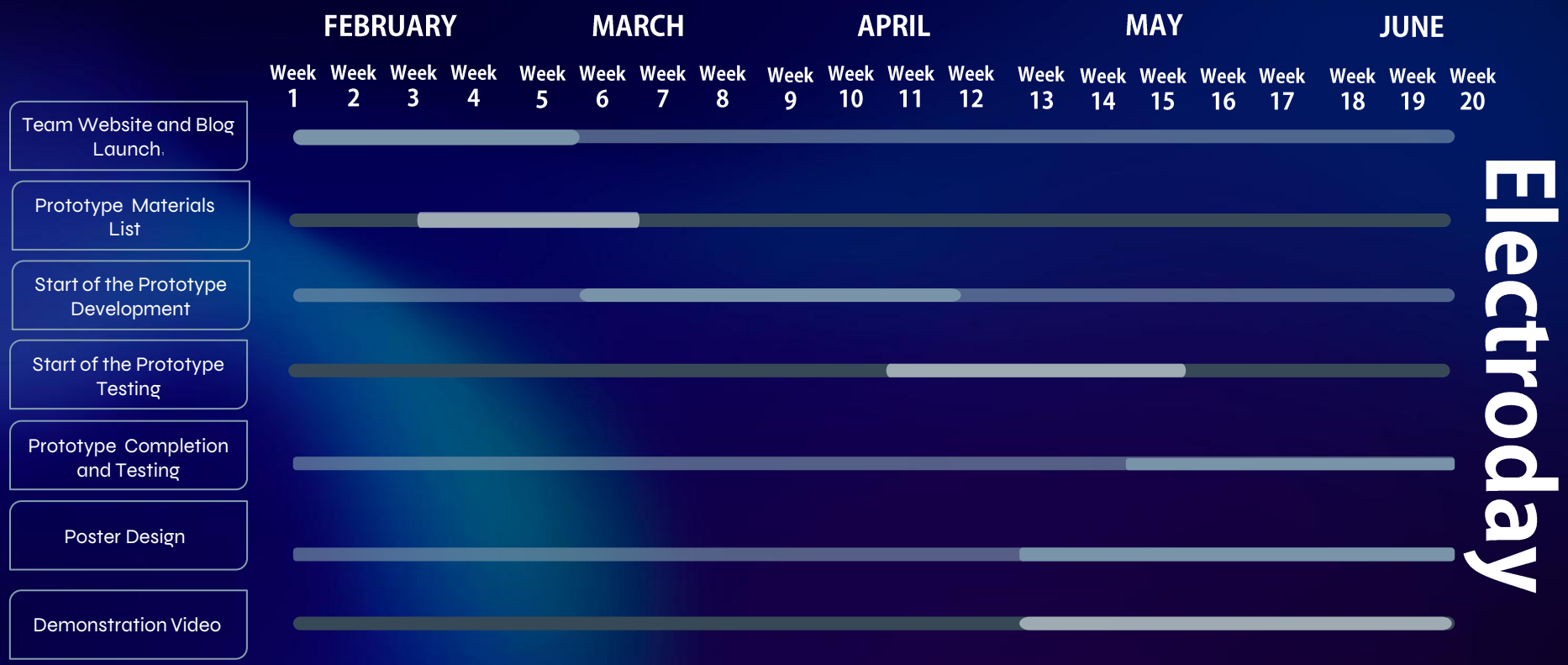

Electroday

Website

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https://web.tecnico.ulisboa.pt/~ist1106726/NAVISENSE/